

```
for (i = 1, s = 0 ; i < 100 ; i++) {  
    s += i*i;  
}
```

```
# s1 = i  
# s2 = s  
li s2, 0  
loop: mul s0, s1, s1  
      add s2, s2, s0  
      addi s1, s1, 1  
      slti s3, s1, 100  
      bnez s3, loop
```

w/ forwarding

Cycles/iteration: 5 instrs + 2
(control hazard) = 7 cycles

700 cycles

LOOP UNROLLING (2 times)

Change to:

```
for (i = 1, s = 0 ; i < 100 ;  
i++) {  
    s += i*i; i++;  
    s += i*i;  
}
```

```
loop:  mul s0, s1, s1
        add s2, s2, s0
        addi s1, s1, 1
        mul s0, s1, s1
        add s2, s2, s0
        addi s1, s1, 1
        slti s3, s1, 100
        bnez s3, loop
```

Cycles/iteration: 8 instrs + 2
(control hazard) = 10 cycles
Total: 500 cycles